

NB41: Feature Selection EDA Validation

SEED=42 | TEST_SIZE=0.2 | 3 panel x k-sweep

0. Hipotez ve Baglam

Danisman EDA: XGBoost feature importance ile MASTER icinde top-200 feature, tum feature lerden daha iyi (train-dagilimi F1: 0.894 vs 0.879). Kuyruk feature lar gurultu mu?

Bizim soru: Bu kazanc, reverse dagilim (%80/20 bootstrap) protokolumuzde de gecerli mi? 3 panelde test ettik.

1. Deney Tasarimi

Her panel icin:

- Stratified %80/20 split (SEED=42)
- M3 missing (is_missing + medyan imputation)
- XGBoost importance ranking (gain)
- Feature subset sweep: k in [10, 25, 50, 100, 200, all]
- BalancedBaggingClassifier (10x LGBM)
- Eval: NB39 protokol (%80/20 bootstrap, N=50, 95% CI)
- Floor reference: F1_floor = 0.333 (prev=0.20)

2. MASTER Sonuclari

k	n_feat	F1_8020	CI_95	MCC	Prec	Rec	vs_floor
10	10	0.4612	[0.355-0.555]	0.307	0.397	0.555	+0.128
25	25	0.5624	[0.469-0.657]	0.442	0.460	0.729	+0.229
50	50	0.5566	[0.460-0.656]	0.434	0.463	0.702	+0.223
100	100	0.5493	[0.450-0.635]	0.427	0.434	0.754	+0.216
200	200	0.5678	[0.488-0.640]	0.455	0.442	0.797	+0.234
all	426	0.5578	[0.472-0.624]	0.441	0.432	0.790	+0.224

2. KANSER Sonuclari

k	n_feat	F1_8020	CI_95	MCC	Prec	Rec	vs_floor
10	10	0.4377	[0.268-0.688]	0.260	0.313	0.750	+0.104
25	25	0.5776	[0.343-0.740]	0.481	0.435	0.877	+0.244
50	50	0.5657	[0.358-0.697]	0.463	0.428	0.853	+0.232
100	100	0.6319	[0.348-0.800]	0.545	0.511	0.847	+0.299
200	200	0.5891	[0.328-0.757]	0.491	0.461	0.837	+0.256
all	502	0.6262	[0.348-0.800]	0.537	0.508	0.837	+0.293

2. PAH Sonuclari

k	n_feat	F1_8020	CI_95	MCC	Prec	Rec	vs_floor
10	10	0.3024	[0.032-0.500]	0.042	0.202	0.620	-0.031
25	25	0.3226	[0.129-0.545]	0.060	0.212	0.700	-0.011
50	50	0.3677	[0.185-0.566]	0.165	0.241	0.813	+0.034
100	100	0.3104	[0.143-0.491]	0.015	0.193	0.800	-0.023
200	200	0.3101	[0.143-0.500]	0.020	0.198	0.747	-0.023
all	408	0.3101	[0.143-0.500]	0.020	0.198	0.747	-0.023

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3. Ana Hipotez Testi: Top-k vs All

Panel	Best	F1	Comp	F1	Diff
MASTER	k=200	0.5678	All	0.5578	+0.0101
KANSER	k=100	0.6319	All	0.6262	+0.0057
PAH	k=50	0.3677	All	0.3101	+0.0576

SONUC: Danisman EDA bulgusu (top-200 > all) MASTER/KANSER/PAH icinde reverse dagilim protokolumuzde de gecerli mi?

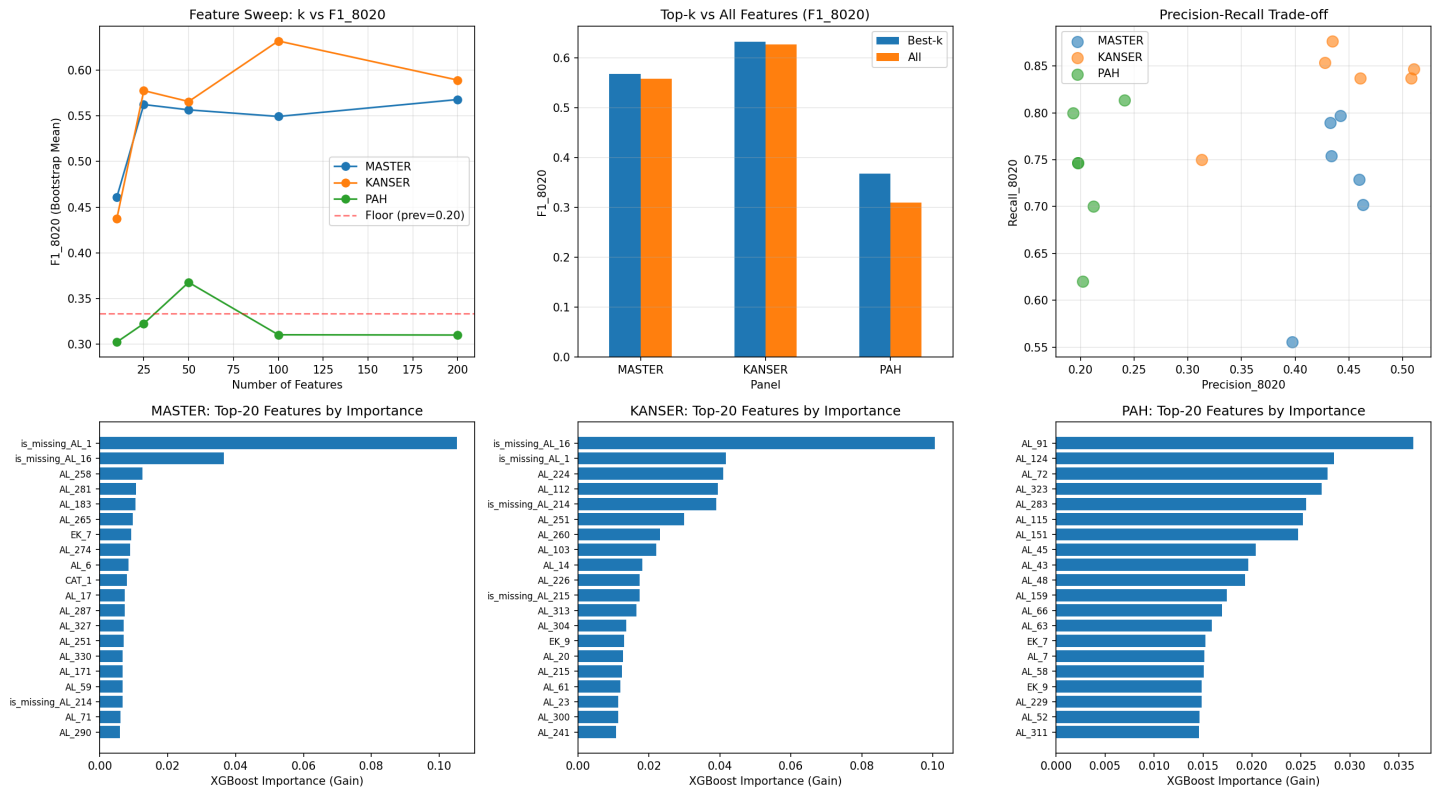
- MASTER: Danisman top-200 onerisini test ettik.
- KANSER: Kucuk panel, top-50/100 avantajli olabilir.
- PAH: Guclu dengesizlik (%17 pos), top-k daha avantajli olabilir.

Floor karsilastirmasi: vs_floor sutununa bak; pozitif ise model trivial baseline i geciyor.

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Gorsel: fig1_ksweep_and_importance.png



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Gorsel: fig2_train_test_gap.png

